Supplementary Materials

Network-Specific Data Partitioning and Sampling

Given the significant differences in the number of edges among different network types across datasets, we further calculated the density of each corresponding network, defined as the ratio of actual edges to the maximum possible edges (see **Table 1**). Based on the density metric, we divided the four types of networks into two groups and applied tailored evaluation strategies to more comprehensively reflect the model's adaptability and performance under different network structures.

For the first group, we used datasets from the STRING, Non-Specific, and LOF/GOF networks. All TF-gene pairs with regulatory relationships were grouped according to their TFs. For each TF, its regulated target genes were randomly split into training and testing sets in a 2:1 ratio, with one-fifth of the training set further held out as a validation set. This TF-based splitting strategy ensures that some regulatory relationships of the same TF appear in the training set while others remain in the testing set, thus more accurately assessing the model's generalization ability to unseen regulatory relationships.

Regarding negative sample construction, all TF-gene pairs not present in the label files were treated as candidate negative samples. To improve the model's discrimination between positive and negative relations, a "hard negative sampling" strategy was employed in the training set: for each positive pair (TFa,Gene_b), a negative pair (TFa,Gene_c) was randomly matched, where Gene_c is not known to be regulated by TFa, ensuring a 1:1 positive-to-negative sample ratio. The validation set adopted the same strategy to randomly match corresponding negative pairs. In contrast, the number of negative samples in the test set was dynamically determined according to the target network's density, making the test set construction better reflect the structural distribution of real regulatory networks. The formula is as follows:

$$\frac{Positive}{Negative} = \frac{NetworkDensity}{1 - NetworkDensity}$$

In the second part, focusing on cell type-specific networks, although the network density is relatively high and there are sufficient positive samples, the number of active transcription factors (TFs) in each network is comparatively small. To ensure the model can fully learn the features of each TF, we split all positive samples of each TF into training, validation, and test sets with proportions of 67%, 10%, and 23%, respectively. Similarly, the potential negative samples corresponding to each TF were divided according to the same proportions, maintaining the integrity of the sample distribution and adapting to the characteristics of high-density networks. This approach facilitates training and evaluation of the model under realistic data distributions.

Supplementary Tables

Table 1 The statistics of four ground-truth networks with TFs and 500 (1000) most-varying genes

Cell types	Cells	STRING				Non-Specific			
		TFs	Genes	Edges	Density	TFs	Genes	Edges	Density
hESC	758	343(351)	511(695)	4257(5149)	0.024(0.021)	283(292)	753(1138)	3441(4617)	0.016(0.014)
hHEP	425	409(414)	646(874)	7523(9003)	0.028(0.024)	322(332)	825(1217)	4129(5351)	0.015(0.013)
mDC	383	264(273)	479(664)	4815(5898)	0.038(0.032)	250(254)	634(969)	3067(3918)	0.019(0.016)
mESC	421	495(499)	638(785)	7762(8479)	0.024(0.021)	516(522)	890(1214)	6893(8030)	0.015(0.013)
mHSC-E	1071	156(161)	291(413)	1371(1826)	0.029(0.027)	144(147)	442(674)	1425(1960)	0.022(0.020)
mHSC-GM	889	92(100)	201(344)	748(1311)	0.040(0.037)	82(88)	297(526)	743(1358)	0.030(0.029)
mHSC-L	847	39(40)	70(81)	137(154)	0.048(0.045)	35(37)	164(192)	279(317)	0.048(0.043)
Cell types	Cells	Specific				LOF/GOF			
		TFs	Genes	Edges	Density	TFs	Genes	Edges	Density
hESC	758	34(34)	815(1260)	4545(7084)	0.164(0.165)	-	-	-	-
hHEP	425	30(31)	874(1331)	9939(15558)	0.379(0.377)	-	-	-	-
mDC	383	20(21)	443(684)	756(1193)	0.085(0.082)	-	-	-	-
mESC	421	88(89)	977(1385)	29,613(42795)	0.345(0.347)	34 (34)	774 (1098)	4169(5742)	0.158(0.154)
mHSC-E	1071	29(33)	691(1177)	11,557(21975)	0.578(0.566)	-	-	-	-
mHSC-GM	889	22(23)	618(1089)	7364(14135)	0.543(0.565)	-	-	-	-
mHSC-L	847	16(16)	525(640)	4398(5180)	0.525(0.507)	-	-	-	-

Table 2 Variants of NP-MCGRN and their modifications.

Model	Euclidean branch	Hyperbolic branch	Spherical branch	Curvature-aware gating	Node-prior decoder
NP-MCGRN	√	√	√	√	√
NP-MCGRN(E+H)	√	√	×	√	√
NP-MCGRN(E+S)	√	×	√	√	√
NP-MCGRN(H+S)	×	√	√	√	√
NP-MCGRN w/o Gate	√	√	√	×	√
NP-MCGRNw/o Prior	√	√	√	√	×

Case study

Table 3 Top 20 TF–gene pairs from three TFs in the hESC 500 dataset

TF	Target	Evidence	TF	Target	Evidence	TF	Target	Evidence
EGR1	TFAP2A	Ground-truth	NANOG	DCLK1	Ground-truth	TFAP2A	JUND	Ground-truth
EGR1	JUND	Ground-truth	NANOG	KLF7	Ground-truth	TFAP2A	SEMA6A	Ground-truth
EGR1	TRAF6	Ground-truth	NANOG	NCAM1	Ground-truth	TFAP2A	HNRNPD	Ground-truth
EGR1	ARFGAP3	Ground-truth	NANOG	PALLD	Ground-truth	TFAP2A	NCAM1	Ground-truth
EGR1	SALL2	Ground-truth	NANOG	DEPDC1B	Ground-truth	TFAP2A	TBX3	Ground-truth
EGR1	PHB2	Ground-truth	NANOG	MAML3	Ground-truth	TFAP2A	SALL2	Ground-truth
EGR1	BCLAF1	Ground-truth	NANOG	ZNF165	Unconfirmed	TFAP2A	WDFY1	Ground-truth
EGR1	RBFOX2	Ground-truth	NANOG	ID1	Ground-truth	TFAP2A	SNRPB	Ground-truth
EGR1	NUP54	Ground-truth	NANOG	TNC	Ground-truth	TFAP2A	MSX1	Ground-truth
EGR1	FOXN3	Ground-truth	NANOG	PTPRZ1	Ground-truth	TFAP2A	FUBP1	Ground-truth
EGR1	SLC19A2	Unconfirmed	NANOG	MECOM	Ground-truth	TFAP2A	SIRT1	Ground-truth
EGR1	PTMA	Ground-truth	NANOG	SEMA6A	Ground-truth	TFAP2A	PALLD	Ground-truth
EGR1	IFI16	Ground-truth	NANOG	C4orf46	Ground-truth	TFAP2A	ENC1	Ground-truth
EGR1	MDM4	Ground-truth	NANOG	RBFOX2	Ground-truth	TFAP2A	EEF1A1	Ground-truth
EGR1	TEAD4	Ground-truth	NANOG	TRAF6	Unconfirmed	TFAP2A	ZNF202	Ground-truth
EGR1	ID1	Ground-truth	NANOG	ETV4	Ground-truth	TFAP2A	RHOBTB3	Ground-truth
EGR1	GABRB3	Ground-truth	NANOG	E2F3	Unconfirmed	TFAP2A	GATAD2A	Ground-truth
EGR1	PLK2	Ground-truth	NANOG	KLHL4	Ground-truth	TFAP2A	PGM2	Ground-truth
EGR1	HNRNPD	Ground-truth	NANOG	AFF4	Ground-truth	TFAP2A	DUSP4	Ground-truth
EGR1	XRCC5	Ground-truth	NANOG	ANKRD28	Ground-truth	TFAP2A	AFF4	Ground-truth

Table 4 Top 20 TF–gene pairs from two TFs in the hESC 500 dataset

TF	Target	Evidence	TF	Target	Evidence
TEAD4	TFAP2A	Ground-truth	OTX2	TFAP2A	Ground-truth
TEAD4	RGPD2	Ground-truth	OTX2	FOXN3	Ground-truth
TEAD4	TRAF6	Ground-truth	OTX2	SOX2	Ground-truth
TEAD4	KIF15	Ground-truth	OTX2	CAMK2D	Ground-truth
TEAD4	POLR2A	Ground-truth	OTX2	PRICKLE1	Ground-truth
TEAD4	AFF1	Ground-truth	OTX2	ZNF8	Unconfirmed
TEAD4	INSIG2	Ground-truth	OTX2	ARHGAP42	Ground-truth
TEAD4	CXXC1	Ground-truth	OTX2	ARHGAP28	Ground-truth
TEAD4	PAWR	Ground-truth	OTX2	YPEL5	Ground-truth
TEAD4	RMI1	Ground-truth	OTX2	ESRP1	Ground-truth
TEAD4	WDR12	Unconfirmed	OTX2	ANKRD13C	Ground-truth
TEAD4	SNRPB	Ground-truth	OTX2	ZNF589	Ground-truth
TEAD4	ASNS	Ground-truth	OTX2	CCPG1	Ground-truth
TEAD4	NFAT5	Ground-truth	OTX2	JUND	Unconfirmed
TEAD4	CEP55	Ground-truth	OTX2	ZNF165	Ground-truth
TEAD4	SIRT1	Ground-truth	OTX2	FAM117B	Ground-truth
TEAD4	PRICKLE1	Ground-truth	OTX2	TNC	Ground-truth
TEAD4	MIA3	Ground-truth	OTX2	SRSF1	Ground-truth
TEAD4	HMGN2	Unconfirmed	OTX2	KAT6A	Ground-truth
TEAD4	ZNF142	Unconfirmed	OTX2	KLHL24	Ground-truth

Hyperparameter tuning

The performance of NP-MCGRN was affected by several key hyperparameters. Among them, node embedding dimension and learning rate had direct effects on representation capacity and training stability. We tuned these hyperparameters on benchmark scRNA-seq datasets constructed from STRING networks, using AUROC and AUPRC as the primary evaluation metrics. All candidate configurations used the same data splits, negative-sample construction strategy and training procedure as the main experiments. The test sets were not used for hyperparameter selection.

We first examined the effect of learning rate on model performance. The candidate learning rates ranged from1×10^{-5} to 1×10^{-3}. As shown in Fig. 1a, NP-MCGRN maintained stable AUROC values across different learning rates. The AUROC varied only slightly across the five STRING datasets, indicating that the model's discriminative ability was not highly sensitive to learning-rate changes. By contrast, AUPRC was more sensitive to the learning rate. Larger learning rates reduced AUPRC, especially on the hESC, hHEP and mHSC-E datasets. This result indicates that overly large parameter updates may weaken the model's ability to rank high-confidence regulatory edges in class-imbalanced GRN inference tasks. In comparison, 1×10^{-5} and 5×10^{-5} achieved more balanced performance on most datasets. Among them, 5×10^{-5} produced the highest mean AUROC and maintained near-optimal AUPRC. Therefore, 5×10^{-5} was used as the default learning rate in subsequent experiments to balance training stability and edge-ranking performance.

We then evaluated the effect of node embedding dimension on model representation capacity. As shown in Fig. 1b, increasing the embedding dimension from 64 to 512 improved AUROC and AUPRC on most datasets. This trend indicates that higher-dimensional latent representations helped capture complex gene regulatory relationships. In particular, 512-dimensional embeddings achieved better overall performance on the mESC and mHSC-E datasets, suggesting that an appropriate increase in representation capacity strengthened the modelling of latent regulatory structures in STRING networks. However, further increasing the embedding dimension to 1024 did not produce continued improvement, and AUPRC decreased on some datasets. This result may indicate that excessively high embedding dimensions introduce redundant representations and increase the model's dependence on the training graph structure, thereby reducing generalization on validation data. Considering both AUROC and AUPRC, 512-dimensional node embeddings showed the most stable performance across the five datasets and were selected as the default embedding dimension for the main experiments.

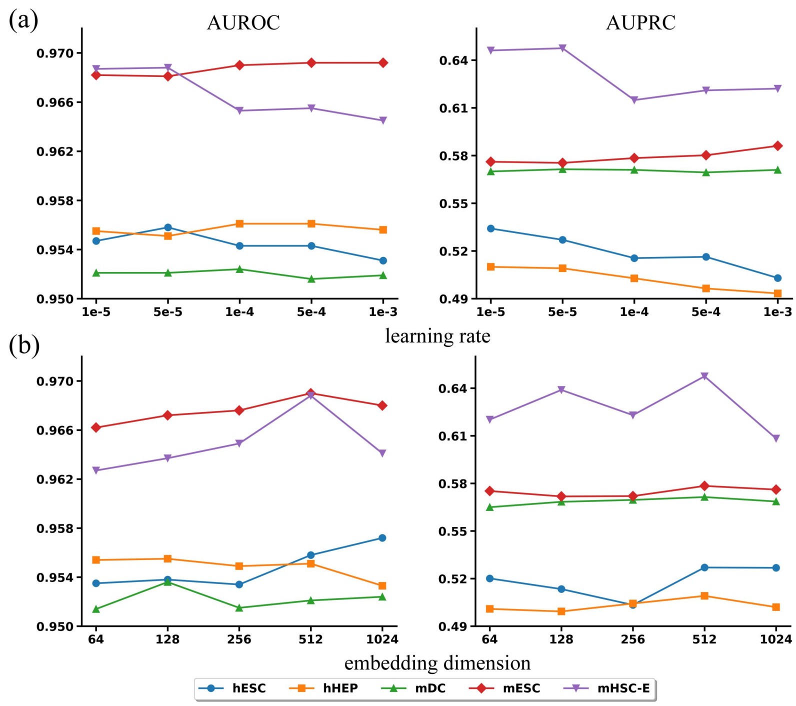

Fig. 1. Hyperparameter analysis results. Effects of learning rate (a) and node embedding dimension (b) on NP-MCGRN performance.

Overall, the hyperparameter tuning results show that NP-MCGRN maintained stable discriminative performance across a relatively wide learning-rate range. Its edge-ranking ability depended more strongly on a conservative learning rate and a moderate representation dimension. The final configuration, with a learning rate of 5×10^{-5} and 512-dimensional node embeddings, achieved a favourable balance between performance and stability across different cell-type datasets. This configuration was therefore used consistently in the subsequent comparison experiments, ablation studies and mechanistic analyses.

Impact of training dataset size on model stability

Training data size affects the representation learning capacity of the model and directly influences its stability in sparse regulatory networks. To evaluate the adaptability of NP-MCGRN under different amounts of training data, we selected four representative datasets: the human hESC and hHEP datasets, and the mouse mHSC-E and mHSC-GM datasets. For each dataset, the model was trained with 10%, 20%, 30%, 40% and 50% of the training samples. AUROC and AUPRC were used to evaluate prediction performance.

As shown in Fig. 2, AUROC and AUPRC increased overall across the four datasets as the proportion of training data increased. For AUROC, hESC increased from 0.7009 with 10% training data to 0.8306 with 50% training data. hHEP increased from 0.8011 to 0.8816, mHSC-E increased from 0.8390 to 0.9103, and mHSC-GM increased from 0.8441 to 0.9115. AUPRC showed the same trend. hESC increased from 0.3575 to 0.5351, hHEP increased from 0.7056 to 0.8131, mHSC-E increased from 0.8674 to 0.9284, and mHSC-GM increased from 0.8490 to 0.9226. These results show that more training samples improved both the discriminative ability and edge-ranking ability of the model. The improvement was especially clear in the hESC dataset, which was more sample-sparse and challenging to predict.

Further comparison across training proportions showed that NP-MCGRN already produced usable predictions with only 10% of the training data. This indicates that the model retained a degree of robustness under low-data conditions. Performance improved markedly as the training proportion increased from 10% to 40%. When the proportion further increased from 40% to 50%, the performance gains became smaller across all four datasets. The mean AUROC increased only from 0.8730 to 0.8835, and the mean AUPRC increased from 0.7892 to 0.7998. These results indicate that NP-MCGRN gradually reached stable performance after receiving a moderate amount of training data. The model did not show excessive dependence on large training sets.

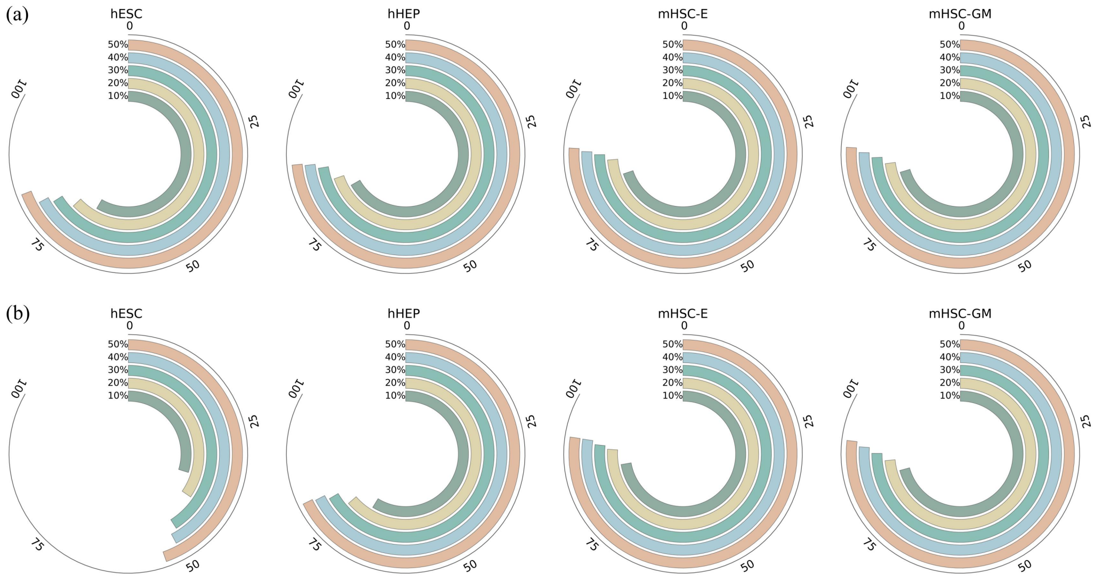

Fig. 2. AUROC (a) and AUPRC (b) performance of NP-MCGRN under different training data sizes.

Overall, the training-data-size experiment showed that NP-MCGRN maintained stable GRN inference performance under different data availability conditions. The model preserved basic predictive ability with a low proportion of training samples. As the amount of training data increased, AUROC and AUPRC further improved and gradually converged. These results indicate that NP-MCGRN can adapt to limited labelled data in sparse GRN settings. With more supervision, it further enhanced its ability to model complex regulatory structures.